

PocketSmart AI

Requirement Analysis

Your Smart Budget & Recommendation Assistant

1. Project Overview

PocketSmart AI is a GenAI-powered budget and recommendation assistant for home decor, party planning and jewelry. It collects user requirements and budget information and provides suitable recommendations.

2. Problem Statement

Users often spend time searching for products and planning events while trying to stay within a fixed budget. PocketSmart AI simplifies this process through budget-aware recommendations.

3. Objectives

- Provide budget-based recommendations.
- Support home decor, party and jewelry planning.
- Reduce search and comparison time.
- Provide an easy-to-use web interface.
- Use Generative AI for personalized recommendations.

4. Target Users

Users who want quick recommendations for home decoration, party planning or jewelry while considering their budget and preferences.

5. Functional Requirements

ID	Requirement	Priority
FR01	User registration	High
FR02	User login and logout	High
FR03	Enter budget and preferences	High
FR04	Generate home decor recommendations	High
FR05	Generate party recommendations	High
FR06	Generate jewelry recommendations	High
FR07	Display recommendation details and estimated cost	High
FR08	Provide fallback recommendations when AI is unavailable	Medium
FR09	Store and display recommendation history	Medium

6. Non-Functional Requirements

- Usability: Simple and understandable interface.
- Performance: Reasonable response time.
- Reliability: Fallback results when AI generation is unavailable.
- Security: Appropriate authentication and session handling.
- Maintainability: Modular routes, services and models.
- Compatibility: Modern desktop and mobile browsers.

7. Software Requirements

Operating System: Windows/Linux/macOS
Programming Language: Python
Backend: FastAPI
Frontend: HTML, CSS, JavaScript, Jinja2
Database: SQLite
AI Service: Google Gemini API
IDE: Visual Studio Code
Version Control: GitHub

8. Hardware Requirements

- Computer or laptop with a modern processor
- Minimum 4 GB RAM recommended
- Internet connection for AI API access
- Keyboard and mouse/touchpad

9. User Stories

ID	User Story
US01	As a user, I want to register and log in so that I can use the application.
US02	As a user, I want to enter my budget so that recommendations match my spending limit.
US03	As a user, I want home decor suggestions based on room type and style.
US04	As a user, I want party suggestions based on event type, guests and budget.
US05	As a user, I want jewelry suggestions based on occasion, type and budget.
US06	As a user, I want to view recommendation details and estimated cost.

10. Requirement Summary

The system requires authentication, three planning modules, budget-aware recommendations, recommendation details and fallback handling. These requirements form the basis for the design, development and testing phases of PocketSmart AI.